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DIC-Net: Upgrade the performance of traditional DIC with Hermite dataset and convolution neural network

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ABSTRACT

Digital image correlation (DIC) is a non-contact optical method that tracks the speckle pattern on specimen surface to calculate the displacement and strain by image correlation algorithm. Although the traditional DIC method can conveniently measure surface deformation, it still has many limitations: (1) the accuracy of displacement and strain calculation needs to be improved in the case of high deformation gradient; (2) under match or over-match can hardly be avoided when the filters are used to reconstruct smooth displacement or strain field, and (3) boundary effect remains unresolved in computing the deformation near the boundary of region of interest or the discontinuous area (e.g. area near crack tip or crack face). Recently, the deep learning based DIC (Deep-DIC) has revealed its attractive ability in handling above issues in traditional DIC, and impressive results have been achieved. The mean value of the absolute error (MAE) on the test set has been optimized to 0.0361 pixels using existing Deep-DIC approaches, which are accompanied by a real-time measurement speed. The network structure and training dataset are two key factors for the deep learning method. However, the current working networks have been modified from other image tasks and cannot fully adapt to the demands of the DIC tasks, and the dataset they generated still has evident flaws, limiting the method's accuracy and generalization performance which is utilized to assess performance on samples outside the training set. In this paper, we firstly proposed a new Hermite dataset that is created by using the high-order Hermite element to take account more complex deformation, then a new network architecture designed for the DIC task has been developed to extract richer deformation features. A test set of 3216 examples containing six different modes of displacement is used to compare the performance of our network with others. The proposed DIC-Net-d achieves the lowest MAE in the test set. Meanwhile, in the Star5 image sets from DIC-Challenge, the proposed DIC-Net-d achieves a spatial resolution of 17.25 pixels and a noise level of 0.0136 which outperforms existing traditional and non-traditional methods. Finally, the strain network trained by our Hermite dataset is also successful in predicting the strain field of Star6 in the DIC challenge. The experiment results show the superiority of the proposed Hermite dataset and new network with respect to other Deep-DIC methods.

1. Introduction

Digital image correlation (DIC) is a non-contact optical measurement method [1], which can obtain displacement field by matching the subsets on the reference image and deformed image [2]. After nearly 40 years of development, DIC has become a flexible and efficient measurement method from the macroscopic to micro- or even nanoscale [3–5]. Traditional DIC methods generally contain the following steps (1) prepare the sample and spray speckles pattern of suitable size and density, (2) record grayscale images of the specimen surface before and after loading, (3) select subset size and shape function for local DIC, calculate the displacement fields using inverse-compositional Gauss-Newton (IC-GN) [6] registration method to minimize specific correlation criterion [7,8], (4) then smooth the displacement fields and calculate strain fields using a global or local strain calculation method [9]. Even though vast efforts have been made to promote the practicality and accuracy, some challenges persist for traditional DIC, they are, (i) optimum subset size and shape function selection is still hard to be achieved [10,11], (ii) boundary effect [11] is obvious due to the split or incomplete subset or smoothing window is used near the crack or the boundary of region of interest [12], and (iii) displacement and strain accuracy is still unsatisfactory under high-gradient deformation cases [13–15]. These problems limit the performance of traditional DIC. Many key factors (such as speckle size, image noise, subset size, shape function, window size, and kernel function within the strain calculation method) strongly affect the accuracy of traditional DIC.

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Abbreviations

x_i	the received signal of neuron
w_i	the neuron weight
f	the activation function of neuron
$\mathcal{D}$	the dataset which contains image pairs and ground truth
$\mathbf{X}_i$	the speckle image pairs
$\mathbf{Y}_i$	the ground truth (GT) of deformation
$\tilde{\boldsymbol{Y}}_i$	the predicted deformation result of encoder-decoder
$\boldsymbol{W}$	the model parameter to be determined
L	loss function which can evaluate the gap between $\tilde{\boldsymbol{Y}}_i$ and $\mathbf{Y}_i$.
N	the total number of samples in dataset
$\mathbf{q}^e$	degrees of freedom (DOFs) at nodes
$\mathbf{N}^e(\mathbf{x})$	Hermite polynomial shape function vector
$\mathbf{q}$	DOFs of all elements
$\mathbf{N}$	global interpolation expression
$\mathbf{u}$	local displacement of element
$\mathbf{u}^c$	assembled full field displacement
$u,\ v$	displacement in x-direction and y-direction
ξ, η	local coordinate of the element
N_i^j	1D quintic Hermite polynomial shape functions
$\bar{N}_i^{(j,k)}$	2D quintic Hermite polynomial shape functions
u_x	the partial derivative of u with respect to x
u_y	the partial derivative of u with respect to y
v_x	the partial derivative of v with respect to x
v_y	the partial derivative of v with respect to y
AEE	the average endpoint error which is used as loss function
u_p, v_p	the predicted result u and v of DIC-Net-d
$u_g,\ v_g$	the ground truth u and v
MAE	the mean value of absolute pixel error
RMSE	the root mean squared error
SR	the spatial resolution
σ^u	the standard deviation of displacement u
α	metrological performance indicator which is product of σ^u and SR
σ^{v_y}	the standard deviation of strain v_y
β	metrological performance indicator which is product of σ^{v_y} and SR

To address the above issues of traditional DIC methods, the convolutional neural network (CNN) [16] in computer vision (CV) may be a promising choice in dealing with the drawbacks of traditional DIC. Convolutional neural network (CNN) is a kind of neural network specially used to process data with grid structure, in which image data can be regarded as two-dimensional pixel grid [17]. In 1989, LeCun Y successfully trained the first self-supervised model with convolution kernel using backpropagation [16]. This builds the foundation for modern CNN. Image categorization is a primary and critical step in CV, it is also the basis for other downstream vision tasks. In the ILSVRC-2012 contest, AlexNet was used to classify the 1.2 million high-resolution images into the 1000 different classes and came out on top with a 15.3% margin of test error [18]. However, the depth of the neural network is limited to about 20 layers, because the deeper network structures would lead to gradient disappearance or gradient explosion, and the network degradation may occur even if the training can converge [19]. Kaiming He et al. presented the well-known ResNet with residual structure in 2015, which solves the difficulty of training deeper networks [20]. In the same year, ResNet won the first prize with a 3.57% test error in the ILSVRC 2015 contest, this made the neural network surpass human in image classification task for the first time. Since then, deep neural networks which use blocks with residual structures as the backbone have been applied for various CV tasks. Deep neural networks have also been applied to the dense prediction task, such as image segmentation [21–23], super-resolution [24,25], depth estimation [26,27], and optical flow estimation [28]. Considering that DIC is also a kind of dense prediction task, we perceive that the CNN can be introduced and form a deep learning-based DIC (Deep-DIC) [29], which may probably achieve success in deformation measurement. Convolution kernel acts as a filter in the image task, and the parameters in the kernel are completely learned from the provided dataset. This makes the deep learning based DIC have the potential to obtain better results than traditional DIC. Meanwhile, the results obtained by CNN have slight boundary effect since no subset is used. Comprehensively considering these factors, we suppose, with the input of image pairs composed of reference and deformed images, the Deep-DIC can predict the displacement end to end well.

There are limited reported works to use CNN to solve DIC tasks. The first work was by Min et al. in 2019 who attempted to combine deep neural networks with DIC for solving strain measurement problems [30]. They modified a fully convolutional network (FCN) [31] to measure strain during tensile testing, however, the structure of FCN is often considered to be inferior to U-Net [22,32]. To build the dataset, Min et al. obtained the input speckle images from a real experiment, and the target displacement fields were accessed by the DIC correlation. Such dataset and network structures make it explicit that the results of the network are unlikely to be better than the traditional DIC approach. More recently in 2021, Boukhtache et al. [33] constructed the Speckle 2.0 dataset and fine-tuned a U-Net of optical flow estimation [28] for the DIC problem. Their trained network (StrainNet-f) finally achieved a mean value of the absolute error (MAE) score of 0.0361 pixels on the test set. Unfortunately, the Speckle 2.0 dataset can only be used to train the displacement network, no strain network was considered. To solve the previous problem, the latest work by Yang et al. proposed a dataset that used the Gaussian function and its partial derivatives to generate the displacement and strain field respectively [29]. The modified image segmentation network [34] was successfully trained to measure deformation. However, the dataset has single deformation mode and low spatial frequency, resulting in weak network's generalization performance.

Above investigations of Deep-DIC show that the dataset with richer deformation modes and the network with more powerful feature extracting ability need to be proposed to achieve a better Deep-DIC model. In this paper, we propose a novel Hermite element based dataset (denoted as Hermite dataset) and a new network (called as DIC-Net) for the DIC task. The Hermite dataset is designed with several parameters that can be easily adjusted to control the magnitude of the displacement and strain field as well as the deformation modes. The DIC-Net is inherited from the U-Net structure [22], while the backbone for feature extraction is comprised of residual blocks and the decoder is also modified to avoid the Checkerboard Artifacts effect [20,35]. In an experimental test set of 3216 samples with six deformation modes, the proposed DIC-Net-d achieves a mean value of the absolute error (MAE) of 0.0130 pixels which is only 48.5% of the existing best one network for DIC. Especially on Star5 image sets from DIC-Challenge, DIC-Net-d has achieved a spatial resolution (SR) of 17.25 pixels with a low noise level of 0.0136 which outperforms existing traditional and non-traditional methods. For strain measurement, DIC-Net-s extends the measurable range of existing network used for DIC. The strain result of Star6 image sets of DIC-Net-s demonstrates a performance that is not inferior to traditional methods.

2. Methodology

2.1. Basic principles of deep learning-based DIC

Neural network has a large quantity of interconnected basic neurons whose structure is simple but learnable [17]. Fig. 1 illustrates a neuron and a neural network which contains three layers. The neuron receives signals $x_i,\ i = 1, 2, 3 \ldots n$ from n other neurons, and these input signals are passed to the current neuron after multiplying by weights $w_i,\ i = 1, 2, 3 \ldots n$. The total input received by the neuron is filtered by the bias w_{n+1}, and finally the output y of the neuron is generated by

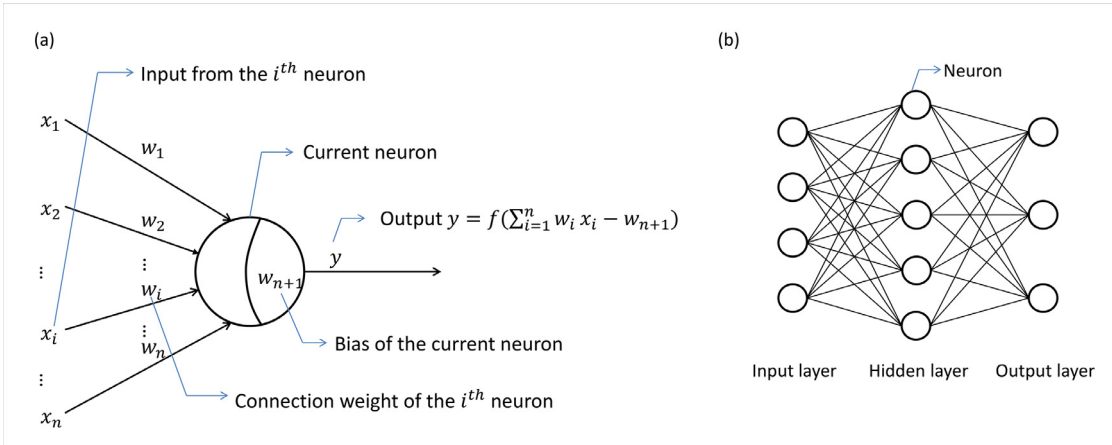

Fig. 1. (a) basic neuron forms the (b) fully connected network which contains only three layers.

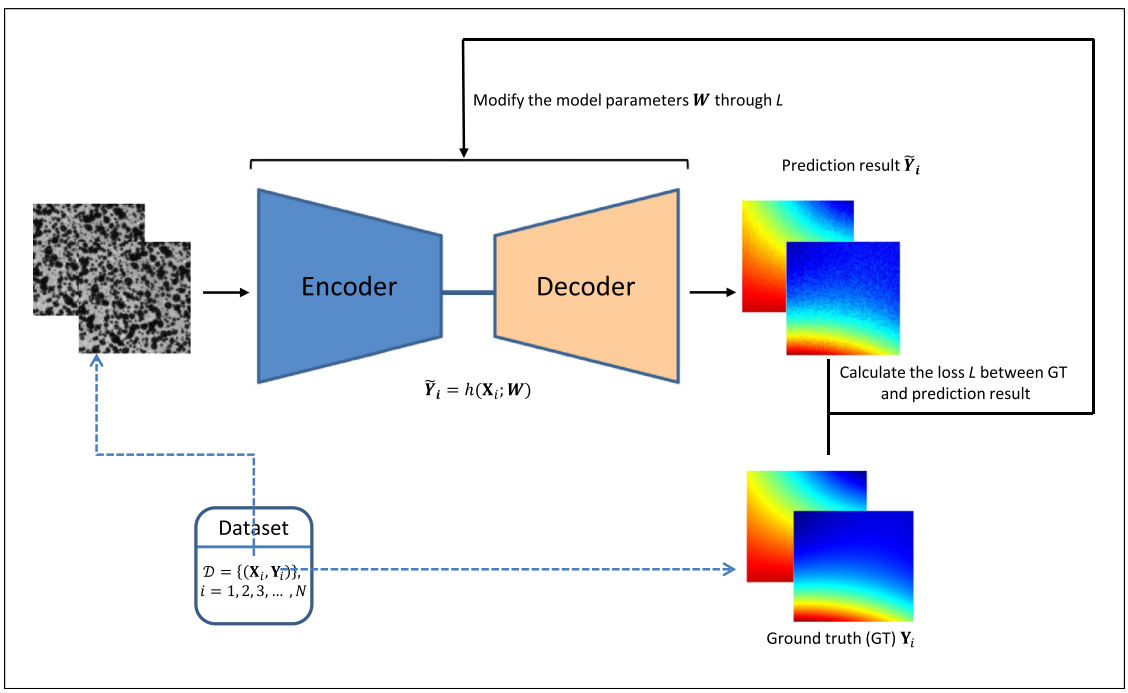

Fig. 2. Overall architecture of Deep-DIC method.

the activation function $y = f(\sum_{i=1}^{n} w_i x_i - w_{n+1})$. The activation function f gives the neural network the ability to map nonlinearly, and ReLU,

$$\mathrm{ReLU}(x) = \begin{cases} 0, & x < 0 \\ x, & x \geq 0 \end{cases}, \tag{1}$$

is often chosen as the activation function in modern neural networks. Fig. 1(b) shows a simple three-layer neural network with fully connected interconnections of neurons between layers, which means that the output of a neuron will be delivered to each neuron in the next layer. The learning process of a neural network is to correct model parameter i.e. the connection weights w_i and bias w_{n+1} of each neuron by the loss of the output and the ground truth. A convolutional neural network (CNN) can be thought of as a change in the connection form of the fully connected network mentioned above, where the output of a neuron is no longer delivered to each neuron but to a specific group of neurons in the next layer [36].

For an end-to-end Deep-DIC model, each pixel point of an image pair is taken as input to each neuron in the input layer, and each neuron in the output layer gives a prediction of the deformation of the corresponding pixel point. To train a CNN that can predict deformation by speckle image pairs, a dataset $D = \{(\mathbf{X}_i, \mathbf{Y}_i)\}$, $i = 1, 2, 3, \ldots, N$ should be defined by a pair of images $\mathbf{X}_i \epsilon\, \mathbb{R}^3$ and the ground truth (GT) of deformation $\mathbf{Y}_i \epsilon\, \mathbb{R}^3$ which contains the deformation in x direction and y direction, the subscript i denotes the label number of the samples in the dataset and N is the total number of samples. An encoder-decoder structure $\tilde{\boldsymbol{Y}}_i = h(\mathbf{X}_i; \boldsymbol{W})$ is constructed by using a CNN, which can be used to implement the mapping from speckle image pair containing spatial-temporal information to the deformation field as Fig. 2 showing, where h is the mapping relationship, $\tilde{\boldsymbol{Y}}_i \epsilon\, \mathbb{R}^3$ is the prediction result, and $\boldsymbol{W}$ is the model parameter to be determined. The learning process can be described as minimizing a loss function L,

$$\min_{\boldsymbol{W}} \sum_{i=1}^{N} L\big(\mathbf{Y}_i, h(\mathbf{X}_i; \boldsymbol{W})\big). \tag{2}$$

Different from the traditional DIC method which solves the deformation field indirectly by minimizing the correlation criterion i.e. ZNSSD [8], the Deep-DIC method directly minimizes the loss between predic-

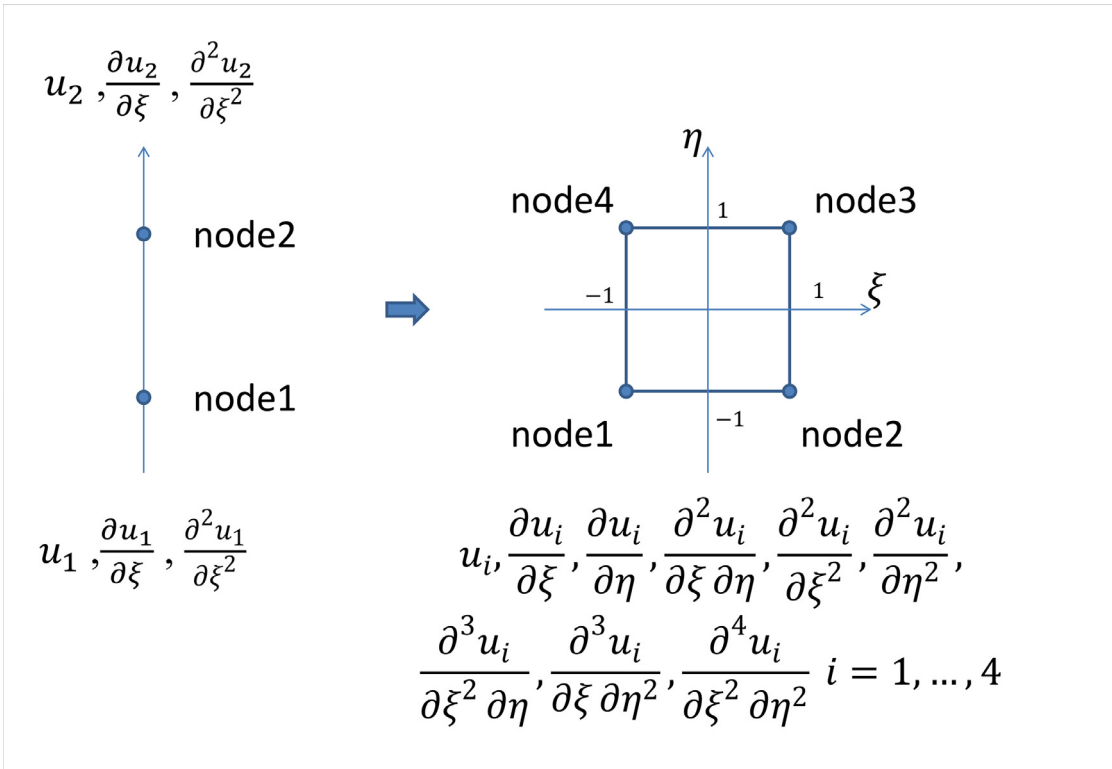

Fig. 3. Definition of the node numbering, nodal degrees of freedom, and local coordinates for 1D and 2D Hermite elements.

tion and ground truth of the deformation, so the data-driven Deep-DIC method has the potential to surpass the traditional method.

2.2. Build the Hermite element dataset

As shown in Section 2.1, the network's ability in predicting deformation is learned from the learning data, therefore the network should be trained using a dataset that is as complete as possible to cope with the complex and variable DIC tasks in real experiments. In previous research, two kinds of competitive datasets for DIC tasks have been proposed.

The first one is Speckle dataset 2.0 designed by Boukhtache et al. [33] which got a desirable result in their work. The StrainNet-f trained by Speckle dataset 2.0 can achieve MAE score of 0.0361 pixels in their test set while having good generalization performance. Despite this, there are still some limitations that limit the performance of the model. Firstly, bicubic interpolation used in Speckle dataset 2.0 can only guarantee C^1 continuity displacement field and the deformation order is low. Secondly, no strain data is considered in Speckle dataset 2.0. Finally, the displacement values on boundary of each sample are set to 0, which disturb the prediction of the boundary point.

Another dataset generated by the 2D gauss function was proposed by Yang et al. to generate both displacement and strain field [29]. They came up with an algorithm to generate their own synthetic speckle image of size 128 × 128 pixels. The displacement field is given by an analytical mixed Gaussian function, after the displacement field is determined, the corresponding strain field can be analytically calculated by taking the spatial derivatives of the displacement field based on the infinitesimal strain assumption. Although such Gaussian function can be used to generate strain field, the network trained by the Gaussian function dataset will adopt a single deformation mode to predict the displacement of the whole field, and the spatial resolution of the predicted results will be higher (lower is better).

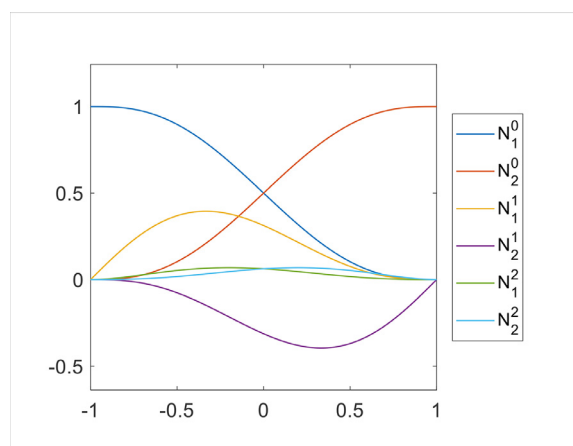

Fig. 4. Hermite polynomial shape function of a 1D two-node Hermite element.

In order to generate both displacement and strain learning data without losing the generalization performance and spatial resolution of the trained model, the Hermite dataset based on Hermite element is proposed to generate displacement fields, which will be applied on the deformed image to create a reference image. The strain ground truth is obtained by differentiating the Hermite shape function. By setting appropriate parameters, two special cases of the Hermite dataset can achieve similar deformation mode as in Speckle 2.0 and Gaussian function datasets.

2.2.1. Hermite finite element in Hermite dataset

The finite element method (FEM) [37] is a powerful numerical tool to solve mechanical problems of complex objects. In FEM, the shape function of the finite element plays the key role in mapping the shape or displacement from the nodes to any point within the element. Firstly, the

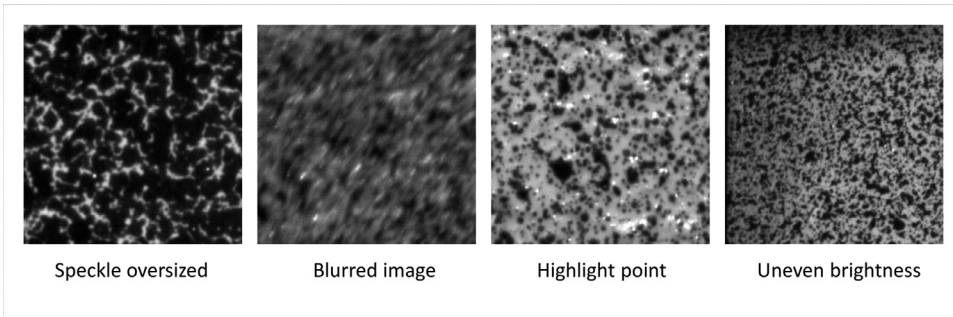

Fig. 5. Different speckle in Hermite dataset.

shape function must contain constant terms and linear terms, since they reflect the properties of rigid body displacement and constant strain, respectively. Secondly, complete polynomials should be selected as much as possible to improve the accuracy of the element. The characteristics of Hermite element and its shape function are suitable for constructing our dataset, because it produces continuous and smooth displacement and even strain on the element boundaries. At the same time, the displacement and strain are degrees of freedom (DOFs) at nodes, which could easily control the deformation within the element. By randomly generating the displacement and strain DOF values on the nodes of the Hermite element, arbitrarily complex full field deformation could be obtained.

The C^2 continuity Hermite element [9,38] is shown in Fig. 3, the two nodes 1D Hermite element contains three DOFs per node to ensure the C^2 continuity, so the nodal displacement of an element $\mathbf{q}^e$ is consist of u_i, $\frac{\partial u_i}{\partial \xi}$, $\frac{\partial^2 u_i}{\partial \xi^2}$, $i = 1, 2$. The six quintic Hermite polynomial shape functions N_i^j, $i = 1, 2$, $j = 0, 1, 2$ corresponding to the two nodes' u_i, $\frac{\partial u_i}{\partial \xi}$, and $\frac{\partial^2 u_i}{\partial \xi^2}$ are drawn in Fig. 4, where the subscript indicates the node number and the superscript denotes the order of the derivative with respect to the local element coordinate $\xi \in [-1, 1]$.

$$\overline{N}_1^0 = -(\xi-1)^3\left(3\xi^2+9\xi+8\right)/16,\ \overline{N}_2^0 = (\xi+1)^3\left(3\xi^2-9\xi+8\right)/16$$
$$\overline{N}_1^1 = -(\xi-1)^3(3\xi+5)(\xi+1)/16,\ \overline{N}_2^1 = -(\xi+1)^3(3\xi-5)(\xi-1)/16$$
$$\overline{N}_1^2 = -(\xi-1)^3(\xi+1)^2/16,\ \overline{N}_2^2 = (\xi+1)^3(\xi-1)^2/16$$

For 2D C^2 continuous Hermite element, it gets a little bit more complicated. To ensure the C^2 continuity of 2D Hermite element, the 9 degrees of freedom at each node are defined as $u_i, \frac{\partial u_i}{\partial \xi}, \frac{\partial u_i}{\partial \eta}, \frac{\partial^2 u_i}{\partial \xi \partial \eta}, \frac{\partial^2 u_i}{\partial \xi^2}, \frac{\partial^2 u_i}{\partial \eta^2}, \frac{\partial^3 u_i}{\partial \xi^2 \partial \eta}, \frac{\partial^3 u_i}{\partial \xi \partial \eta^2}, \frac{\partial^4 u_i}{\partial \xi^2 \partial \eta^2}$ $i = 1, \ldots, 4$. All the 36 Hermite polynomial shape functions are obtained as products of the above 1D shape functions,

$$\begin{aligned} \tilde{N}_i^{(j,k)}(\xi,\eta) &= \mathrm{N}_{i'}^j(\xi)\mathrm{N}_{i''}^k(\eta) w^j h^k, \\ i &= 1, \ldots, 4, \\ i', i'' &= 1, 2, \\ j, k &= 0, 1, 2, \end{aligned} \tag{3}$$

where $N_i^{(j,k)}(\xi,\eta)$ is associated with the degree of freedom $(\partial^{j+k} u/\partial x^j \partial y^k)_i$ of node i of the rectangular element, w and h are the half width and half height of element respectively, i' and i'' refer to the node number in corresponding 1D element and they take values of 1 or 2 if the local coordinate of node i equals −1 or 1.

Then the local displacement $\mathbf{u}(\mathbf{x})$ of point at x inside of element could be expressed as,

$$\mathbf{u}(\mathbf{x}) = \mathbf{N}^e(\mathbf{x})\mathbf{q}^e, \tag{4}$$

after the displacement within each element determined, the full field displacement $\mathbf{u}^c$ can be assembled using the following formula,

$$\mathbf{u}^c = \sum_{e=1}^{E} \mathbf{N}^e(\mathbf{x})\mathbf{q}^e = \mathbf{N}\mathbf{q} \tag{5}$$

where e is the label number of element and E is total number of elements i.e. $\mathbf{N}^e(\mathbf{x})$ and $\mathbf{q}^e$ denote the shape functions and DOFs of element number e, respectively. $\mathbf{q}$ is the DOFs of all elements, $\mathbf{N}$ is the global interpolation expression.

For short, the dimension and continuity of the selected Hermite element determine the mode of the displacement field. In [9,38], Hermite element is used to smooth the noisy displacement field and compute strain field by differentiation, $\mathbf{q}$ is computed by regularization method. In this paper, $\mathbf{q}$ of all Hermite elements are randomly generated within controlled range, which results in a controllable deformation of each element. Although 2D C^2 continuous Hermite element is used as an example here, actually one can get an nD C^m continuous displacement field by switching shape function, where n is the dimension and m is the continuity.

2.2.2. Generation of Hermite dataset

Displacement and strain field generation

As suggested in the paper [39], in order to facilitate the acquisition of the displacement field located in the reference image coordinate system, the displacement field should be applied to the deformed image to obtain the reference image. So the first step in generating the Hermite dataset is to prepare some speckle images as deformed images. A total of 429 deformed images are used in dataset, among which the most speckle images (256 × 256) of various qualities are obtained from real experiments. Fig. 5 shows some poor quality speckle images of the dataset which are due to speckle spraying or image capture mistake. The speckle images from real experiments can improve the ability of network in handling practical measurements. Then the images are further cropped to 193 × 193 pixels and the element size is chosen to be 5 × 5, 9 × 9, 17 × 17, 33 × 33, 65 × 65 pixels, to make the displacement field values generated by Hermite element all be on the integer pixel.

With the deformed image prepared, the reference image will be obtained from the deformed one [39,40]. Displacement field used to restore the reference image is randomly generated using following steps,

(1) Select one element size from numbers 5, 9, 17, 33 and 65 pixels, which will generate different displacement spatial frequencies.
(2) Split the each deformed image of size 193 × 193 pixels using the selected element size.
(3) The nodal DOF $\mathbf{q}^e = [u_i, \frac{\partial u_i}{\partial \xi}, \frac{\partial u_i}{\partial \eta}, \frac{\partial^2 u_i}{\partial \xi \partial \eta}, \frac{\partial^2 u_i}{\partial \xi^2}, \frac{\partial^2 u_i}{\partial \eta^2}, \frac{\partial^3 u_i}{\partial \xi^2 \partial \eta}, \frac{\partial^3 u_i}{\partial \xi \partial \eta^2}, \frac{\partial^4 u_i}{\partial \xi^2 \partial \eta^2}]$ $i = 1, \ldots, 4$ is randomly generated. Range of random displacement degree and strain degree (conforming to uniform distribution) varies according to element size, and different parameter setting strategies are chosen for training the strain and displacement networks as shown in Table 1.For higher displacement gradients, $\frac{\partial^2 u_i}{\partial \xi \partial \eta}, \frac{\partial^2 u_i}{\partial \xi^2}, \frac{\partial^2 u_i}{\partial \eta^2} \in [-0.001, 0.001]$, $\frac{\partial^3 u_i}{\partial \xi^2 \partial \eta}, \frac{\partial^3 u_i}{\partial \xi \partial \eta^2} \in [-0.0001, 0.0001]$, $\frac{\partial^4 u_i}{\partial \xi^2 \partial \eta^2} \in [-0.00001,\ 0.00001]$ for all element size.
(4) Express the local displacement field of all element as $\mathrm{u}(\mathbf{x}) = \mathbf{N}^e(\mathbf{x})\mathbf{q}^e$.
(5) Assemble to get full field, $\mathbf{u}^c = \sum_{e=1}^{E} \mathbf{N}^e(\mathbf{x})\mathbf{q}^e = \mathbf{N}\mathbf{q}$, calculate the gradient $u_x = \sum_{e=1}^{E} \frac{\partial \mathbf{N}^e(\mathbf{x})}{\partial \boldsymbol{x}} \mathbf{q}^e$, so as the u_y, v_x, v_y.
(6) Output: $\mathbf{u}^c$, u_x, u_y, v_x, v_y.

Table 1
The random range of nodal DOF.

Network	Element size	u_i	$\frac{\partial u_i}{\partial \xi}$	$\frac{\partial u_i}{\partial \eta}$
Displacement network	5	[−0.5,0.5]	[−0.03,0.03]	[−0.03,0.03]
	9	[−0.5,0.5]	[−0.03,0.03]	[−0.03,0.03]
	17	[−1,1]	[−0.03,0.03]	[−0.03,0.03]
	33	[−1.5,1.5]	[−0.03,0.03]	[−0.03,0.03]
	65	[−2,2]	[−0.03,0.03]	[−0.03,0.03]
Strain network	5	[−0.1,0.1]	[−0.03,0.03]	[−0.03,0.03]
	9	[−0.2,0.2]	[−0.03,0.03]	[−0.03,0.03]
	17	[−0.5,0.5]	[−0.03,0.03]	[−0.03,0.03]
	33	[−1.5,1.5]	[−0.03,0.03]	[−0.03,0.03]
	65	[−2,2]	[−0.03,0.03]	[−0.03,0.03]

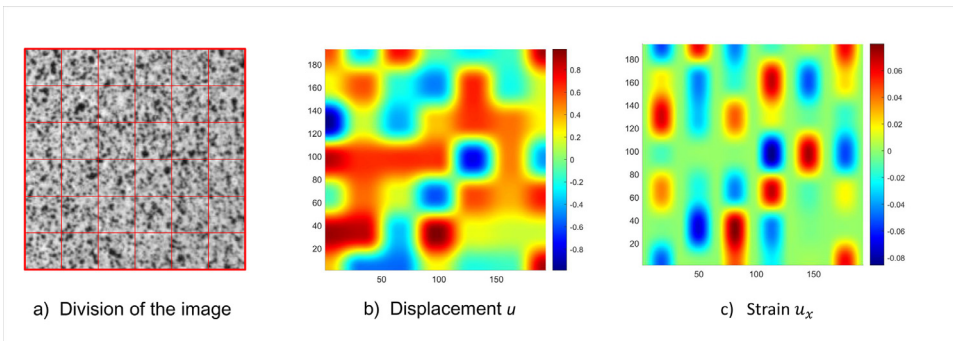

Fig. 6. Displacement and strain field generated by element with size of 33, (a) Division of the 193 × 193 image by the element of size33, (b,c) displacement u and strain u_x.

The displacement and strain field generated by element size of 33 pixels are shown in Fig. 6 below. Similar results could also be obtained by other element sizes with different spatial frequencies. Here it could be found that the Hermite dataset is going to have the same effect as the Gaussian function dataset and Speckle dataset 2.0, if element size is set to 128 and the shape function is replaced with 2D C^1 continuity, respectively. In other words, the effect of these two existing datasets is just two special cases of Hermite dataset. So the models trained by using Hermite dataset may have better generalization performance.

Hermite dataset generation

Before discussing how to generate the dataset, it is important to emphasize that for the DIC task, the dataset should contain a pair of images composed of a reference image and a deformed image, as well as the ground truth of displacement and strain field used for the displacement network and strain network respectively. In addition, the image sizes used in this paper are all 128 × 128.

To generate enough data to train the network and keep from overfitting, we performed a hundred different transformations on each of the 429 images, resulting in a total of 42,900 samples. Among them, 38,600 samples are used as the training set to train the network, and the remaining 4300 samples are used as the test set to test the generalization performance of the network. Every deformed image (193 × 193) will go through the following steps;

(1) Generate 20 random displacement fields (193 × 193) of each of the five size elements (5, 9, 17, 33, 65) as the section above shown.
(2) The generated random displacement field is applied to the deformed image to produce 100 reference images (193 × 193) with Gaussian noise level no more than 0.5% added.
(3) Get two integers x and y randomly in the range [1,66], generate point (x, y). This process shown in Fig. 7 is repeated a hundred times, creating a hundred different square regions with (x, y), $(x, y + 127)$, $(x + 127, y + 127)$, $(x + 127, y)$ as the corner point.
(4) Use the square regions to crop the generated reference image, deformed image, displacement field, strain field (193 × 193) to 128× 128.

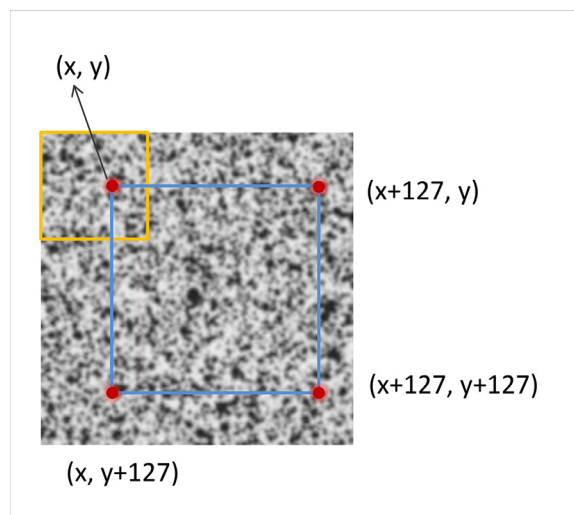

Fig. 7. Random generation of a 128× 128 size image from the size of 193× 193.

In Section 2.2, based on theory of the Hermite finite element, to generate the Hermite dataset three parameter should be selected, i) random range of $\mathbf{q}^e$, ii) 2D C^2 continuous shape function, and iii) square element with five different sizes selected from 5, 9, 17, 33 and 65. In the later experiments, our displacement network and strain network use the dataset with these parameters. The statistics of the maximum strain magnitude of the strain network's training set are shown in the Fig. 8 below. It is obvious, these parameters can be modified to generate more complex fields by increasing the random range of $\mathbf{q}^e$, switching to higher-order shape function or using non-square elements.

2.3. Proposed DIC-Net

In this study, the network structure maintains the design of U-Net, however, the encoder and decoder have been adapted for the character-

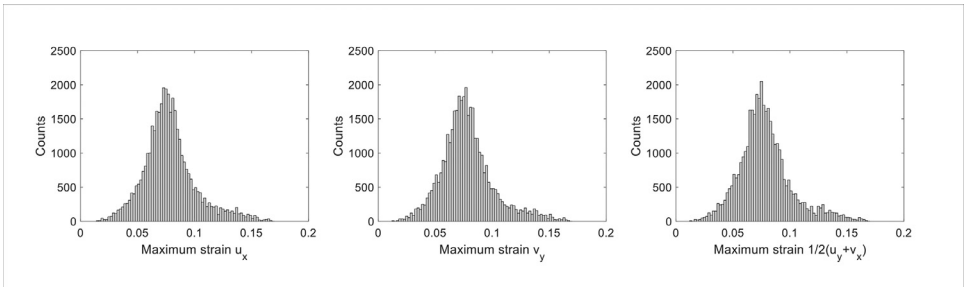

Fig. 8. Statistics of the strain network's training set with 38,600 samples.

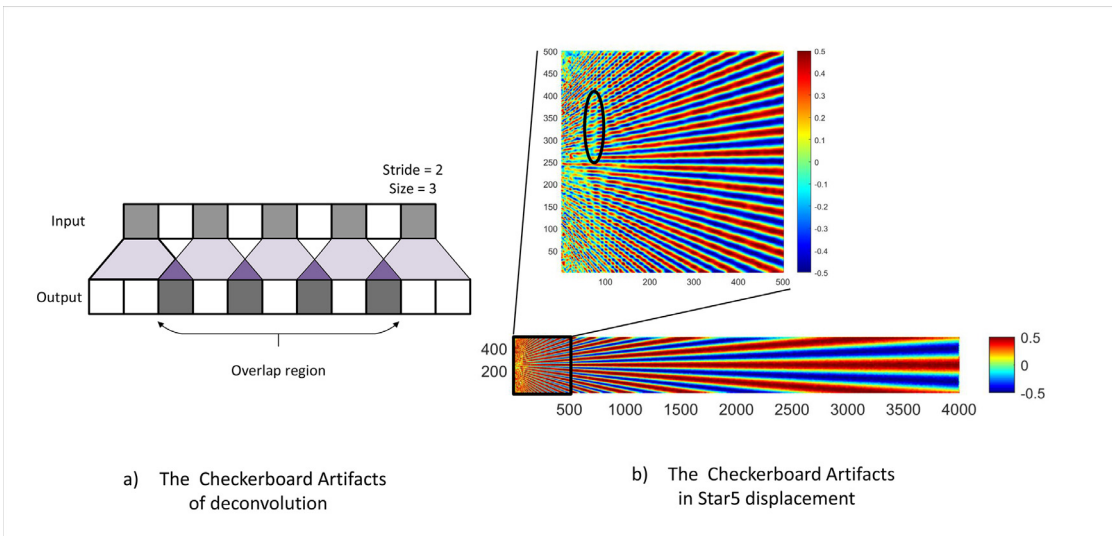

Fig. 9. Demonstration of Checkerboard Artifacts: (a) procedure of deconvolution which rises to Checkerboard Artifacts, (b) Checkerboard Artifacts produced by StrainNet-f.

istics of the DIC task. In the stage of encoder, two adjustments are applied. First, the residual block is used to construct a deeper network for extracting more features. Second, the relative position extractor (RPE) is proposed for equal-size transformation of the input grayscale images. In the stage of decoder, the deconvolution operation is discarded, because deconvolution will induce Checkerboard Artifacts if the size of the convolution kernel is not divisible by the stride size [35]. Checkerboard Artifacts are due to the uneven overlap which results in repeated superpositions of values at some points in the output tensor. We found that the Checkerboard Artifacts occur in the displacement field of Star5 image set [41] predicted by StrainNet-f [33] shown in Fig. 9.

As shown in Fig. 10, the proposed network (named DIC-Net) takes a pair of grayscale images as input, and outputs the displacement field or strain field directly. Different from the other two networks that extract features from the grayscale image directly, DIC-Net firstly transforms the gray information into an equal-size feature map through the relative position extractor (RPE). RPE simply achieves such an effect by a convolution operation with a kernel size of 3 and a stride size of 1, besides, batch normalization and Relu activation are also used in RPE. Then, six residual blocks are applied to extract richer features and expand channels to 1024. To avoid Checkerboard Artifacts, in the stage of decoder, upsample block which contains a bilinear upsample operation and two convolution operations with a kernel size of 3 and a stride size of 1 is used to replace deconvolution. As same as the original U-Net architecture, the feature extracted in the encoder is concatenated to every upsample block's output. In the end, the network restores the displacement and strain field through two convolution operations with kernel size of 3 and 1 respectively.

Considering that the strain to be calculated is actually the gradient of the displacement, the features of strain network to be extracted from the input are similar to displacement prediction. The architecture of DIC-Net is also used for predicting strain field. In the following paragraphs, the networks used to predict displacement and strain are marked as DIC-Net-d and DIC-Net-s, respectively.

2.4. Training and predicting

The following average endpoint error (AEE) [33] is used as loss function to train our network.

$$\mathrm{AEE} = \frac{\sum_{i=1}^{H}\sum_{j=1}^{W}\sqrt{\left(u_p(i,j)-u_g(i,j)\right)^2+\left(v_p(i,j)-v_g(i,j)\right)^2}}{HW}, \tag{6}$$

where H and W are the height and width of the output, u and v are the displacement field in x-direction and y-direction, and the subscripts p and g represent the result of the DIC-Net and the ground truth, respectively. For the training of the strain network DIC-Net-s, u and v in Eq. (6) are replaced with the strain ε_x, ε_y, and ε_{xy}.

Training of DIC-Net

DIC-Net is implemented on the PyTorch (version 1.10.0) platform with the following training strategies,

(1) The learning rate is kept constant at 0.0001, and although the loss would decrease a little more if the learning rate decreases with the increasing of epoch, such strategy is chosen to prevent overfitting.
(2) The batch size of mini-batch gradient descent is set to 16 to balance the memory consumption of GPU and the fastest seeking path.

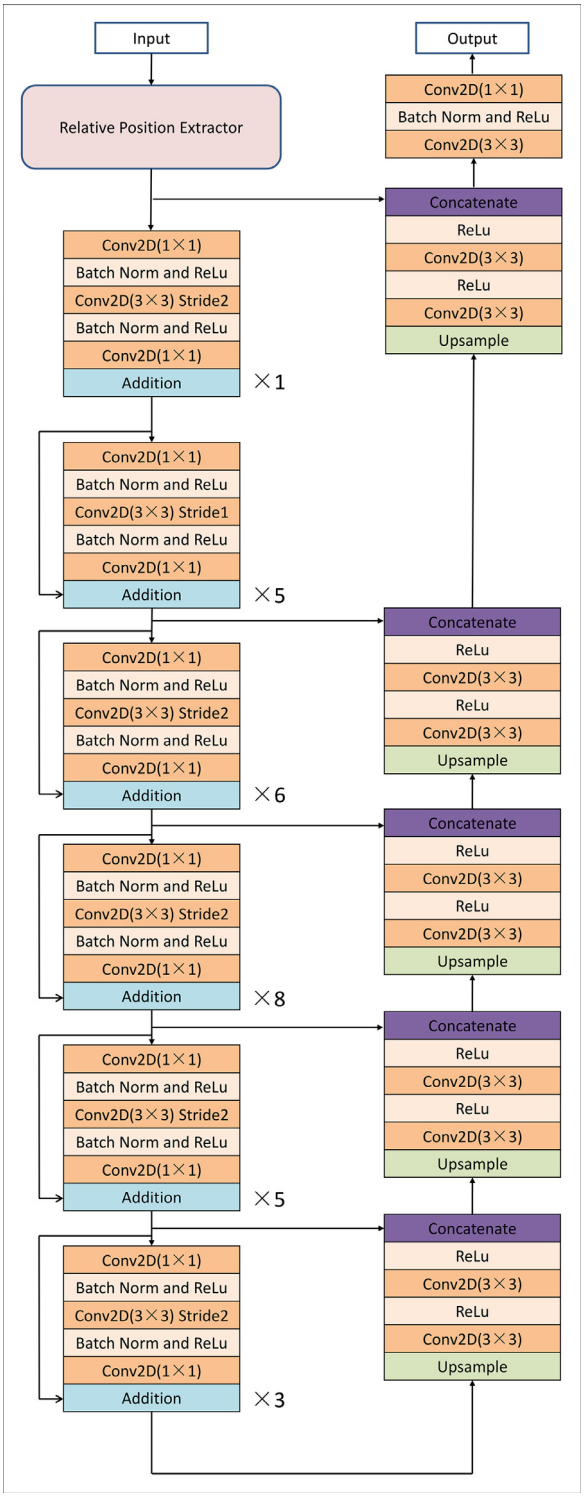

Fig. 10. Architecture of the proposed DIC-Net.

(3) Adam [42] is chosen as the optimization method, the hyper-parameters β_1 and β_2 are set to 0.9 and 0.999.

The training result of DIC-Net-d and DIC-Net-s are shown in Table 2 below.

Predicting using DIC-Net

The trained DIC-Net can perform the end-to-end deformation prediction. With a pair of speckle images as input, the DIC-Net will directly output the displacement field or strain field. The input grayscale image will be transformed into a tensor of size $1 \times 2 \times H \times W$, due to sequential order of reference image and deformed image during training and the downsampling layer numbers, the input needs to follow the following rules,

Table 2
Performance of the DIC-Net in the training set and test set.

AEE/pixel	DIC-Net-d	DIC-Net-s
Training set	0.0357	0.0111
Test set	0.0339	0.0101

(1) The number 1 in the first dimension means that only one pair of images is input.
(2) The number 2 in the second dimension indicates that there are two images, where the deformed image is in front and the reference image is behind.
(3) H and W are the height and width of the image, respectively. The input for predicting does not need to be as large as the training size nor does it need to be equal in height and width. H and W only need to be an integer multiple of 32, for example, 64×64, 64×128, and 512×4096 pixels are all valid inputs.

In real experiments, the size of the input is not controllable. When using DIC-Net, the input image can be expanded to the closest valid size by means of padding, which has little effect on the prediction of the result since there is slight boundary effect in this approach.

3. Experiments and discussion

This section, the experimental test set with 3216 samples (image pairs) is prepared, including 785 displacement fields of pure translations Eq. (7), 357 linear displacement fields Eq. (8), 714 quadratic displacement fields Eq. (9), 445 displacement fields generated by Gaussian function Eq. (10), 497 periodic displacement fields Eq. (11) generated by sinusoidal functions, and 418 displacement fields Eq. (12) generated by simulating strain concentrations. The maximum displacement magnitude of all displacement fields is limited to about one pixel, finally, 0 to 1.18% random Gaussian noise is added to all samples,

$$\begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} a_1 \\ b_1 \end{bmatrix}, \tag{7}$$

$$\begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} a_1 & a_2 & a_3 & a_4 \\ b_1 & b_2 & b_3 & b_4 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ x \\ y \\ xy \end{bmatrix}, \tag{8}$$

$$\begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} a_1 & a_2 & a_3 & a_4 & a_5 & a_6 & a_7 & a_8 & a_9 \\ b_1 & b_2 & b_3 & b_4 & b_5 & b_6 & b_7 & b_8 & b_9 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ x \\ y \\ xy \\ x^2 \\ y^2 \\ x^2y \\ xy^2 \\ x^2y^2 \end{bmatrix}, \tag{9}$$

$$\begin{bmatrix} u \\ v \end{bmatrix} = \sum_{j=1}^{N} \begin{bmatrix} \frac{a_x^j}{2\pi\sigma_{x_0}^j\sigma_{y_0}^j} e^{-\frac{1}{2}\left(\left(\frac{x-x_0^j}{\sigma_{x_0}^j}\right)^2+\left(\frac{y-y_0^j}{\sigma_{y_0}^j}\right)^2\right)} \\ \frac{b_y^j}{2\pi\sigma_{y_1}^j\sigma_{x_1}^j} e^{-\frac{1}{2}\left(\left(\frac{x-x_1^j}{\sigma_{x_1}^j}\right)^2+\left(\frac{y-y_1^j}{\sigma_{y_1}^j}\right)^2\right)} \end{bmatrix}, N = 2, \tag{10}$$

$$\begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} asin\left(\sigma_{x_0}(x-x_0)+\sigma_{y_0}(y-y_0)\right) \\ bsin\left(\sigma_{x_1}(x-x_1)+\sigma_{y_1}(y-y_1)\right) \end{bmatrix}, \tag{11}$$

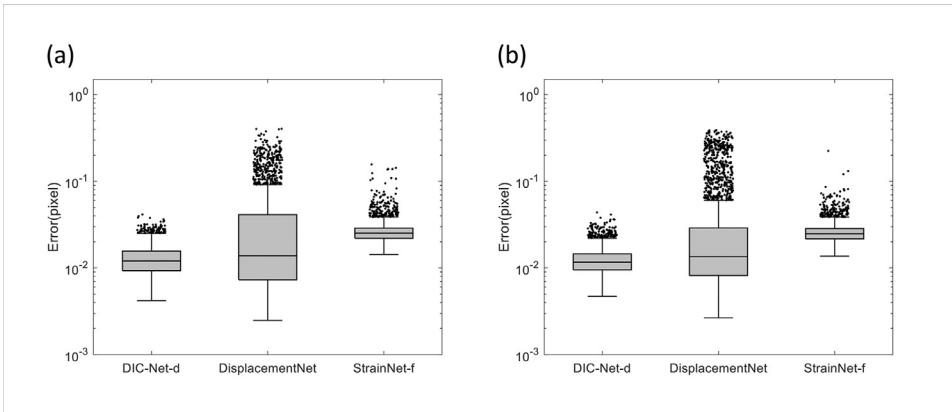

Fig. 11. MAE boxplots of the displacement predicted by the three networks: (a) MAE distribution of displacement in x direction, (b) MAE distribution of displacement in y direction.

Table 3
Prediction error statistics of three networks on 3216 samples.

Displacement mode (number of samples)	Network	Average MAE/pixel		Average RMSE/pixel	
		u	v	u	v
Pure translations (785)	DIC-Net-d	**0.0124**	**0.0115**	**0.0155**	**0.0147**
	DisplacementNet	0.0182	0.0154	0.0197	0.0172
	StrainNet-f	0.0265	0.0255	0.0442	0.0418
Gaussian (445)	DIC-Net-d	**0.0128**	**0.0122**	**0.0161**	**0.0156**
	DisplacementNet	0.0171	0.0164	0.0199	0.0194
	StrainNet-f	0.0256	0.0265	0.0441	0.0433
Periodic displacement (497)	DIC-Net-d	**0.0157**	**0.0169**	**0.0201**	**0.0217**
	DisplacementNet	0.1191	0.1539	0.1395	0.1783
	StrainNet-f	0.0288	0.0295	0.0467	0.0458
Strain concentrations (418)	DIC-Net-d	**0.0137**	**0.0122**	**0.0175**	**0.0157**
	DisplacementNet	0.0410	0.0252	0.0529	0.0315
	StrainNet-f	0.0301	0.0280	0.0504	0.0457
Linear displacement (357)	DIC-Net-d	**0.0120**	**0.0117**	**0.0151**	**0.0148**
	DisplacementNet	0.0169	0.0125	0.0187	0.0144
	StrainNet-f	0.0254	0.0251	0.0435	0.0408
Quadratic displacement (714)	DIC-Net-d	**0.0121**	**0.0117**	**0.0153**	**0.0149**
	DisplacementNet	0.0143	0.0152	0.0165	0.0174
	StrainNet-f	0.0250	0.0243	0.0427	0.0405
Total (3216)	DIC-Net-d	**0.0130**	**0.0126**	**0.0165**	**0.0161**
	DisplacementNet	0.0356	0.0378	0.0417	0.0440
	StrainNet-f	0.0268	0.0263	0.0450	0.0427

Note: the numbers in bold face indicates the lowest error among three networks under comparison.

$$\begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} asin\left(\frac{x}{x_0}\right)sin\left(\frac{y}{y_0}\right) \\ bsin\left(\frac{x}{x_1}\right)sin\left(\frac{y}{y_1}\right) \end{bmatrix}, \tag{12}$$

where {u, v} are the displacement in x and y direction, coefficient {a, b, a_n, b_n, a_x^j, b_y^j}, {σ_{x_0}, σ_{x_1}, $\sigma_{x_0}^j$, $\sigma_{y_0}^j$, $\sigma_{x_1}^j$, $\sigma_{y_1}^j$} and {x_0, y_0, x_1, y_1} are randomly generated to control the center and magnitude of the displacement field, {x, y} are the coordinates of each pixel in the reference image.

To evaluate the accuracy of the prediction results, the following mean value of absolute pixel error (MAE) and root mean squared error (RMSE) are used to calculate the gap between the result and the ground truth,

$$\text{MAE} = \frac{1}{H \cdot W} \sum_{i=1}^{H} \sum_{j=1}^{W} \left| u_p(i,j) - u_g(i,j) \right| \tag{13}$$

and

$$\text{RMSE} = \sqrt{\frac{\sum_{i=1}^{H} \sum_{j=1}^{W} \left(u_p(i,j) - u_g(i,j) \right)^2}{H \cdot W}}. \tag{14}$$

3.1. Comparison between DIC-Net-d and other networks

The experimental test set mentioned in the above paragraphs contains various modes of displacement, which is used in the experiments of this section to evaluate the generalization ability and accuracy of the networks. The parameters of StrainNet-f [33] and DisplacementNet [29] trained by Boukhtache et al. are downloaded from [43,44] to compare with DIC-Net-d. It is worth pointing out that none of the displacement modes of the experimental test set used are in the training set of DIC-Net, but the maximum displacement magnitude are within the range of all three networks' training set. Therefore this experimental test set is fair for the three Deep-DIC methods.

The displacement fields of 3216 samples are predicted using three networks which are trained with different dataset, i.e. DIC-Net-d, DisplacementNet, and StrainNet-f, respectively. The box plot shown in Fig. 11 with y-axis in logarithmic coordinates illustrates the MAE distribution of the predicted results. In Table 3, the mean values of MAE and RMSE in x direction and y direction are tabulated for quantitative evaluation, the errors on the six displacement modes are listed separately, the last row of Table 3 shows the total errors.

In terms of accuracy, the proposed DIC-Net-d achieves the best results on both average RMSE and average MAE. The average RMSE is 0.0165 pixels in x direction which is 39.6% of the previous best Dis-

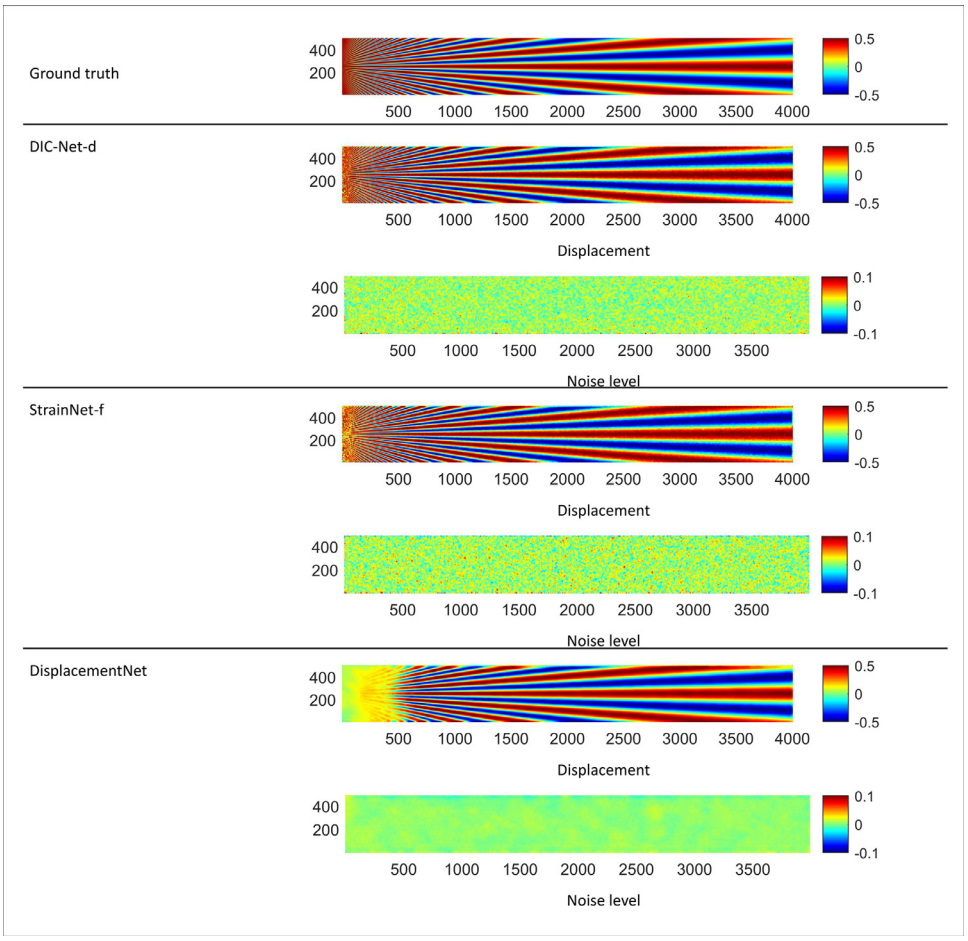

Fig. 12. Star5 displacement and noise level predicted by three networks.

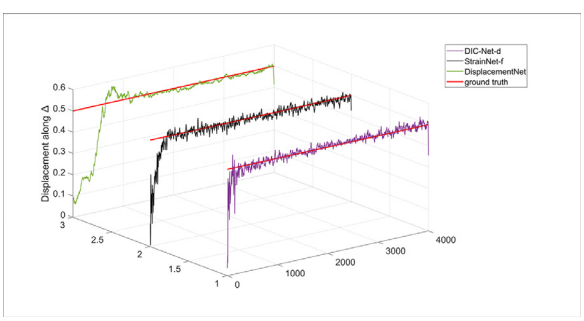

Fig. 13. Displacement of Star5 along the horizontal axis of symmetry Δ per column in pixel.

placementNet. The average MAE is 0.0130 pixels in x direction which is 48.5% of the previous best StrainNet-f. The similar results are reached in the y direction. In terms of generalization performance, both DIC-Net-d and StrainNet-f maintain good consistency without dramatic fluctuations on different displacement modes. On the other hand, DisplacemenNet has a inferior performance on the periodic displacement fields (Eq. (11)) and strain concentrations (Eq. (12)). Especially in the mode of periodic displacement fields, the MAE and RMSE of DisplacemenNet exceed 0.1 pixels. However, excluding these two displacement modes, the accuracy of DisplacemenNet is still worse to that of DIC-Net-d but better than that of StrainNet-f.

Part of the success of DIC-Net-d is due to the fact that the Hermite dataset is interpolated using a polynomial of fifth order, which provides a higher accuracy compared to the bicubic interpolation used in Speckle 2.0. To enable the network to learn the higher accuracy and more complex deformations provided by the Hermite dataset, DIC-Net-d applies a deeper downsampling layers with residual blocks which is accompanied with more powerful feature extraction ability. The method using float precision images of training set in [33] is abandoned and the addition of noise makes the network learn to filter noise to enhance the robustness of the model.

3.2. Spatial resolution and metrological performance indicator

In this section, the Star5 image set [41] from DIC-Challenge 2.0 is utilized to compare the spatial resolution (SR) of DIC-Net-d, DisplacemenNet, and StrainNet-f. As defined in [33,45], the ground truth of Star displacement values along the center horizontal axis of symmetry Δ is 0.5 pixels constantly, but the measured one tends to be lower when approaching the low displacement spatial resolution region on the left side. The spatial resolution is defined as period of the vertical sine displacement whose amplitude is equal to 0.45 pixels. The predicted displacement along Δ will be fitted by a 12th order polynimal to calculate the SR (lower one is better) for avoiding the disturbance of local fluctuation.

In addition, a model that can capture lower spatial resolution of displacement will be more sensitive to noise. This means that it is contradictory to obtain a lower SR and a lower noise level for conventional methods. Therefore, a metric is needed to comprehensively evaluate the performance of the method. The standard deviation σ^u is calculated from the displacement predicted by the noisy and noiseless Star images to quantify the noise level. As described in [33,45], the product of σ^u and SR denoted by α and named "metrological performance indicator" could be used to evaluate our network.

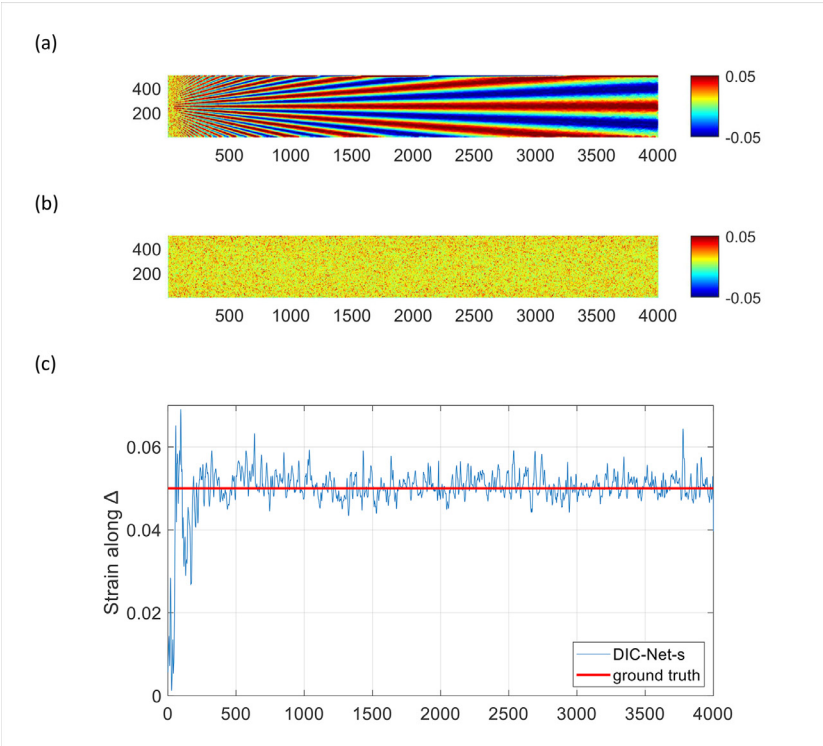

Fig. 14. Strain field of Star6 predicted by DIC-Net-s, (a) result of full field strain of Star6, (b)noise level, (c) the predicted strain along the center horizontal axis of symmetry Δ.

Table 4
The performance of three networks in Star5 image set.

Network	SR	σ^u	α
DIC-Net-d	**17.25**	0.0136	0.234
StrainNet-f	26.93	0.0163	0.439
DisplacemeNet	52.66	**0.0041**	**0.216**

Note: the numbers in bold face indicates the best one among three networks under comparison.

Figs. 12 and 13 show the results predicted by the proposed DIC-Net-d and other two competitive networks (StrainNet-f and DisplacementNet), the SR, σ^u and α values are listed in Table 4. DisplacementNet has a low noise level σ^u of 0.0039 but sacrifices the ability to predict displacements with low SR. The SR of DisplacementNet is 52.66 pixels which is because the dataset used to train DisplacementNet lacks samples with high spatial frequency displacement. Compared with StrainNet-f, the proposed DIC-Net-d simultaneously improves performance in terms of spatial resolution and noise level which achieves a SR of 17.25 pixels and noise level σ^u of 0.0136. The StrainNet-f achieves the SR of 26.93 pixels and noise level σ^u of 0.0163. It is noted that although DisplacemeNet and DIC-Net-d have similar α value, the construction method of Gaussian dataset for DisplacementNet makes it difficult to adjust the spatial frequency of the training data to balance the SR and noise levels, because the deformation generated by the Gaussian function tends to be more smooth. For DIC-Net-d and StrainNet-f, which are both capable of resolving displacement with low SR, DIC-Net-d outperforms StrainNet-f in terms of both SR and noise level. The α of DIC-Net-d is 0.234, which is far better than 0.439 of StrainNet-f.

3.3. Measurement of strain field by DIC-Net-s

The Hermite dataset with the parameters in Section 2.2.2 is applied to train the DIC-Net-s successfully. Due to the limitation of the magnitude in the training set, the maximum displacement of input is expected to be less than about 2 pixels. For the strain of input images, the predictable range is shown in Fig. 8. Therefore the Star6 image set (maximum displacement v about ± 2.3 pixels and maximum strain υ_y of ± 0.05) [45] from DIC-Challenge 2.0 is chosen as the validation sample for our DIC-Net-s.

To the best of our knowledge, existing work with approach of deep learning has not been able to measure the strain whose deformation is accompanied with high spatial frequency and the strain magnitude exceeds 0.02. The proposed DIC-Net-s trained with Hermite dataset extends the application of deep learning-based DIC. For this reason, the results of proposed DIC-Net-s in this section will be discussed in comparison with the strain measured by conventional subset-based local DIC. The first- and second order shape function are applied respectively, the subset size 11 and 21 pixels are also used for comparison. After displacement field is calculated by conventional DIC, the point-wise least square [46] using the first-order kernel function and 5 by 5 smoothing window size is utilized to compute strain field. The predicted strain field $\boldsymbol{v_y}$ of Star6 and the strain along the center horizontal axis of symmetry Δ are shown in Fig. 14. Spatial resolution and noise level σ^{υ_y} are still calculated as defined in Section 2.2. The SR of strain measurement is defined as period of the vertical sine strain whose amplitude is equal to 0.045, and noise level σ^{υ_y} is the standard deviation calculated from the strain field predicted by the noisy and noiseless Star6 images. The "metrological performance indicator" β for strain is defined as the squared of the spatial resolution times the noise level which is different from the one for displacement [45]. All results are listed in Table 5 and comparison of the strain along Δ is shown in Fig. 15.

The result shows that DIC-Net-s can resolve the strain of Star6 image set well. Among the methods for comparison, DIC-Net-s achieved the best SR of 22.09 pixels, but with the highest noise level of 0.0098. For conventional DIC, the results provided here is consistent with [45]. The conventional DIC with the parameters selected in this section all have a larger SR than DIC-Net-s, however they are all better than DIC-Net-s in terms of noise level. Comprehensive consideration from the "metrological performance indicator" β, DIC-Net-s with a β of 4.78 is close to the DIC (subset size of 11 and 2nd order shape function) with a β of 4.11.

Table 5
The strain measurement spatial resolution and noise level by different methods.

Method		SR	Noise level	β
DIC-Net-s		**22.09**	0.0098	4.78
DIC	$11-1^{st}$	47.07	0.0034	7.53
	$11-2^{nd}$	22.96	0.0078	4.11
	$21-1^{st}$	89.24	**0.0012**	9.56
	$21-2^{nd}$	33.75	0.0028	**3.19**

Note: The $\mathbf{mm}-\boldsymbol{n^{xx}}$ used in the column "Method" indicates that DIC with subset size mm and $\boldsymbol{n^{xx}}$ order subset shape function.

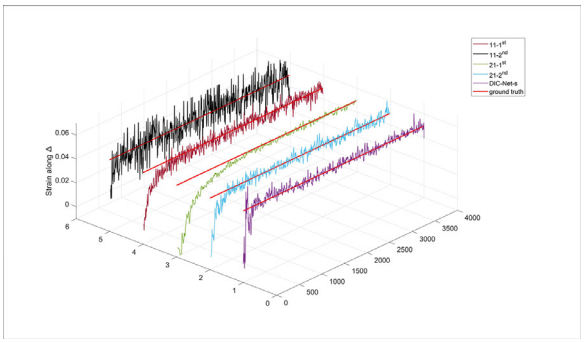

Fig. 15. Comparison of the strain along Δ between DIC-Net-s and traditional methods.

Compared with displacement prediction, the strain prediction is more challenging, the same noise level has much greater impact on strain. In addition, when using displacement network for prediction, the input images only needs to satisfy the condition that the maximum displacement does not exceed the range of the training set, however for strain prediction, the range that can be predicted is limited by both the strain and displacement magnitude in the training set. There is still much potential for improving performance of strain prediction by adjusting the parameter of the Hermite dataset and the corresponding network structure.

4. Conclusions

In this paper, our works confirm the feasibility of deep learning based DIC for deformation measurement, and the proposed DIC-Net further improves the accuracy, generalization and spatial resolution of state-of-the-arts. The main conclusions of this paper are as follows;

(1) The proposed Hermite dataset provides a method of generating displacement and strain field datasets simultaneously for deep learning based DIC. Due to the property that the deformation magnitude could be controlled conveniently, the 2D C^2 Hermite shape functions of the element can produce variable deformation modes, which give the network powerful generalization performance and good spatial resolution.

(2) DIC-Net is designed to extract the features in Hermite dataset while being able to handle noise. The experiments demonstrate that DIC-Net-d has more powerful feature extraction capability by deepening the downsampling layer, while outperforming existing networks in terms of accuracy and generalization performance. In the experimental test set, DIC-Net-d achieves an impressive MAE score of 0.0130 pixels and 0.0126 pixels in the x direction and y direction which is only 48.5% and 47.9% of the existing best one that has been trained with a different type of image dataset.

(3) DIC-Net-s successfully measures the strain of high spatial frequency and extends the range of strain magnitude that the existing deep learning based DIC has measured, which is benefited from the attractive property of Hermite dataset.

The experiment results and the prediction performance on Star5 image set (spatial resolution of 17.25 and a noise level of 0.0136) illustrate that a well-designed deep learning based DIC is able to outperform traditional methods. However, more efforts are needed to build networks that can predict displacements and strains over larger deformation ranges. At the same time, handling the noise in strain prediction is still a challenge for DIC-Net.

Data available

The MATLAB and pytorch code of Hermite dataset generation algorithm and the DIC-Net will be released at the URL: https://github.com/YinWang20/DIC-Net.

Declaration of Competing Interest

We declare that we have no financial and personal relationship with other people or organizations that can inappropriately influence our work, there is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the position presented in, or the review of, the manuscript entitled "DIC-Net: Upgrade the Performance of Traditional DIC with Hermite Dataset and Convolution Neural Network".

Acknowledgment

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